

Exp No: 13

EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS AND STORE THE 32-BIT RESULT IN INTERNAL RAM

OBJECTIVES

- To write an Embedded C program to add an array of 16-bit numbers.
- To obtain and store the result as a 32-bit value.
- To store the 32-bit result in the internal RAM of the AT89C51 microcontroller.

PROGRAM

```
#include <reg51.h>

void main(void)
{
    unsigned int i;
    unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
    unsigned long sum = 0;
    for (i = 0; i < 5; i++)
    {
        sum = sum + array[i];
    }

    P0 = (unsigned char)(sum & 0xFF); // Lower 8 bits
    P1 = (unsigned char)((sum >> 8) & 0xFF); // Next 8 bits
    P2 = (unsigned char)((sum >> 16) & 0xFF); // Upper 8 bits

    while(1); // keep output stable
}
```

OUTPUT

The screenshot displays the uVision IDE interface for a project named "Experiment14VADD_ARRAY.uvproj". The main window shows the disassembly of the C code, which is being executed. The code defines a main function that calculates a sum of an array and outputs the result to three parallel ports (P0, P1, P2).

Registers:

Register	Value
r0	0x12
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x01
r7	0x00
sys	0x00
a	0x00
b	0x00
sp	0x01
wp_ptr	0x33
wp	0x00
PC	0x0080C
status	554
sec	0.00027700
pwr	0x00

Disassembly:

```
15: void main(void)
16: {
17:     unsigned int i;
18:     sum = 0;
19:
20:     unsigned char result[4] _at_ 0x30;
21:     void main(void)
22:     {
23:         unsigned int i;
24:         sum = 0;
25:         for(i = 0; i < 5; i++)
26:         {
27:             sum = sum + array[i];
28:         }
29:         P0 = (unsigned char)(sum & 0xFF);
30:         P1 = (unsigned char)((sum >> 8) & 0xFF);
31:         P2 = (unsigned char)((sum >> 16) & 0xFF);
32:         P3 = (unsigned char)((sum >> 24) & 0xFF);
33:         result[0] = sum & 0xFF;
34:         result[1] = (sum >> 8) & 0xFF;
35:         result[2] = (sum >> 16) & 0xFF;
36:         result[3] = (sum >> 24) & 0xFF;
37:     }
38: }
```

Parallel Port 0:

Port	7	6	5	4	3	2	1	0
P0	0	0	0	0	0	0	0	0

Parallel Port 1:

Port	7	6	5	4	3	2	1	0
P1	0	0	0	0	0	0	0	0

Parallel Port 2:

Port	7	6	5	4	3	2	1	0
P2	0	0	0	0	0	0	0	0

Command:

```
WS 1, 'sum
WS 1, 'array[0]
WS 1, 'array[1]
WS 1, 'array[2]
WS 1, 'array[3]
WS 1, 'array[4]
```

Watch 1:

Name	Value	Type
sum	0x001ABCC	ulong
array[0]	0x1111	uint
array[1]	0x2222	uint
array[2]	0x8888	uint
array[3]	0x4444	uint
array[4]	0xABCD	uint

Simulation Status: t1: 2803.42990200 sec L19 C:1 CAP. NUM. SCRL. OVR. R/W